ELEC 3TP3 - Cheat Sheet

Amaraam Ul Haque

Signal Types

- Analog - all values possible + cont.
  - ↳ set of ∞ # of discrete values
- Digital - not all values possible + cont.
  - ↳ set of limited # of discrete values
- Analog & discrete - f(t) only available @ certain times
  - ↳ sampling
- Digital & discrete - basically computerized values
- Periodic - $f(t) = f(t+T_0)$, for all t
- Causal - $[0, \infty)$
- Non-causal - $[-\#, \pm\#]$
- ↳ Anti causal - $[-\infty, 0)$

Basic Operations

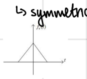
- Shift: $\varphi(t) = f(t-T)$
  - ↳ +T = time delay
  - ↳ −T = time advance
- Scale: $\varphi(t) = f(at)$
  - ↳ $|a|>1$ = compression
  - ↳ $|a|<1$ = expansion
- Both: $\varphi(t) = f(at-b)$
  - ↳ shift b, then scale a
  - ↳ scale a, then shift b/a

Energy & Power

- $E_f = \int_{-\infty}^{\infty} |f(t)|^2 \, dt$
- $P_f = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 \, dt$

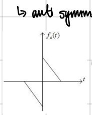
Signal Types

- $u(t)$ - unit step: $[0, \infty]$
  - $f(t) = u(t-x) \to [x, \infty)$
  - $f(u(t-x) - u(t-y)) \to [x, y]$
  - $f(u(t+x) - u(t+y)) \to [-x, -y]$
- $\delta(t)$ - unit impulse / direct delta
  - restricts area under graph
  - $\int_{-\infty}^{\infty} \delta(t)\, dt = 1$ → unit area
  - $\delta(t) = 0$, for $t \neq 0$
  - ↳ $\varphi(t)\delta(t) = \varphi(0)\delta(t)$
  - ↳ $\varphi(T)\delta(t-T) = \varphi(t)\delta(t-T)$
  - ↳ $\frac{d}{dt} u(t) = \delta(t)$
  - ↳ $\int \delta(t)\, dt = u(t)$
- Sinusoid - $f(t) = \cos(\omega t)$
  - ↳ ω = freq = rad/s
  - ↳ $f = \frac{\omega}{2\pi}$ freq = Hz
  - ↳ $T = \frac{1}{f}$ = period
- Complex Exponential
  - $f(t) = \underbrace{e^{j\omega t}}_{} = \cos(\omega t) + j\sin(\omega t)$
  - ↳ magnitude = 1
  - ↳ ωt = frequency
- Even $f_e(t) = f_e(-t)$
  - ↳ symmetric w/ respect to t=0



$A = \int_{-a}^{a} f_e(t)\, dt = 2\int_0^a f_e(t)\, dt$

- Odd $f_o(t) = -f_o(-t)$
  - ↳ anti symmetric w/ respect to=0



$A = \int_{-a}^{a} f_o(t)\, dt = 0$

- Odd Even Components:

$f(t) = \frac{1}{2}\underbrace{[f(t) + f(-t)]}_{\text{even}} + \frac{1}{2}\underbrace{[f(t) - f(-t)]}_{\text{odd}}$

Linear Systems

- Response = $y(t)$ = output due to initial conditions + output due to input
- Total response = zero input response + zero state response
  - $y[x(t)] = y[0]$ $y[x_{t=0}(t)]$
- zero input response - $0 = P(D)\,x(t) = Q(D)\,y(t)$

$$0 = \underbrace{(D^N + a_1 D^{N-1} + \dots + a_{N-1}D + a_N)}_{\Sigma = 0} y_0(t)$$

$$D^N y_0(t) = c\lambda^N e^{\lambda t}$$

$$\vdots$$

$$0 = \cancel{c}(\lambda^N + a_1\lambda^{N-1} + \dots a_{N-1}\lambda' + a_N)\cancel{e^{\lambda t}}$$

$$0 = \lambda^N + a_1\lambda^{N-1} \dots a_{N-1}\lambda' + a_N$$

$$0 = Q(\lambda)$$

has N roots ⟶ $Q(\lambda) = (\lambda-\lambda_1)(\lambda-\lambda_2)\dots(\lambda-\lambda_N)$ ⟶ characteristic equation

↳ $\lambda_1 \dots \lambda_N$ - eigenvalues / characteristic roots / natural freq. of the system

+ solutions: → $y_0(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t} + \dots + c_N e^{\lambda_N t}$

* $c_1, c_2 \dots c_N$ from given $y(0), y'(0) \dots$

- Repeated 'r' roots: $0 = (D-\lambda)^r y_0(t)$
  - modes: $e^{\lambda t}, te^{\lambda t}, t^2 e^{\lambda t}, \dots t^{r-1} e^{\lambda t}$
- Complex roots: complex eigenvalues λ_i need complex conjugate c_i → ∵ system is real, ∴ the #'s are real too

$\lambda_1 = \alpha + j\beta$
$\lambda_2 = \alpha - j\beta$ ⟶ $y_0(t) = c_1 e^{(\alpha+j\beta)t} + c_2 e^{(\alpha-j\beta)t}$

Convolution

$y(t)$ = input * impulse response

$= x(t) * h(t)$

$= \int_{-\infty}^{\infty} x(\tau)\cdot h(t-\tau)\, d\tau$

- $x(t) * \delta(t) = x(t)$

Convolution Properties

- commutative: $A*B = B*A$
- distributive: $A*[B+C] = A*B + A*C$
- associative: $A*(B*C) = (A*B)*C$
- shift: $x_1(t-T_1) * x_2(t-T_2) = y(t-T_1-T_2)$

Graphical Convolution

$y(t) = x(t) * h(t)$

1) $x(t)$ & $h(t)$ → $x(\tau)$ & $h(\tau)$
2) mirror: $h(-\tau)$
3) shift: $h(t-\tau)$
4) find common domain
5) integrate over domain

ℰ Operator

$\mathcal{E}\, x[n] = x[n+1]$
$\mathcal{E}'\, x[n] = x[n+1]$
$\mathcal{E}^2\, x[n] = x[n+2]$
$\mathcal{E}^{-1}\, x[n] = x[n-1]$

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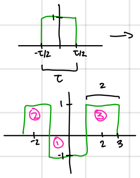


⟶ $P_\tau(\omega) = \tau \operatorname{sinc}\left(\frac{\omega\tau}{2\pi}\right)$

$\underline{\tau = 2:}$

$p_1 = -P_\tau(t)$ ⟶ $p_1(t) \leftrightarrow -P_\tau(\omega)$
$p_2 = P_\tau(t+2)$ ⟶ $p_2(t) \leftrightarrow P_\tau(\omega)\cdot e^{+j\omega 2}$
$p_3 = P_\tau(t-2)$ ⟶ $p_3(t) \leftrightarrow P_\tau(\omega)\cdot e^{-j\omega 2}$

$x(t) = p_1(t) + p_2(t) + p_3(t)$

$X(\omega) = -P_\tau(\omega) + P_\tau(\omega)\cdot e^{j\omega 2} + P_\tau(\omega)\cdot e^{-j\omega 2}$

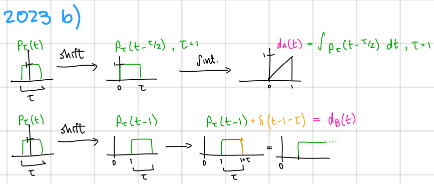
$P_\tau(\omega) = \tau\operatorname{sinc}\left(\frac{\omega\tau}{2\pi}\right)$
↳ $\tau = 2$

$= -2\operatorname{sinc}\left(\frac{\cancel{2}\omega}{\cancel{2}\pi}\right) + 2\operatorname{sinc}\left(\frac{\cancel{2}\omega}{\cancel{2}\pi}\right)\cdot e^{2j\omega} + 2\operatorname{sinc}\left(\frac{\cancel{2}\omega}{\cancel{2}\pi}\right)\cdot e^{-2j\omega}$

$= +2\operatorname{sinc}\left(\frac{\omega}{\pi}\right)\left[-1 + e^{2j\omega} + e^{-2j\omega}\right]$

$= 2\operatorname{sinc}\left(\frac{\omega}{\pi}\right)\left[-1 + 2\cos(2\omega)\right]$

2023 b)



$d(t) = d_A(t) + d_B(t) = \underbrace{\int p_\tau(t-\tau/2)\, dt}_{d_A(t)} + \underbrace{p_\tau(t-1)}_{d_{B1}(t)} + \underbrace{\delta(t-1-\tau)}_{d_{B2}(t)}$, where $\tau = 1$

$p_\tau(t) \to P_\tau(\omega)$
$p_\tau(t-\tau/2) \to P_\tau(\omega)\cdot e^{-j\omega\tau/2}$
$\int p_\tau(t-\tau/2)\, dt \to P_\tau(\omega)\cdot e^{-j\omega\tau/2}\cdot\frac{1}{j\omega}$
$d_A(t) \to D_A(\omega)$

$D_A(\omega) = P_\tau(\omega)\cdot e^{-j\omega\tau/2}\cdot\frac{1}{j\omega}$

$p_\tau(t) \to P_\tau(\omega)$
$p_\tau(t-1) \to P_\tau(\omega)\cdot e^{-j\omega 1}$
$\to P_\tau(\omega)\cdot e^{-j\omega}$
$d_{B1}(t) \to D_{B1}(\omega)$

$D_{B1}(\omega) = P_\tau(\omega)\cdot e^{-j\omega}$

$\delta(t) \to 1$
$\delta(t-1-\tau) \to 1\cdot e^{-j\omega 1}\cdot e^{-j\omega\tau}$
$\to e^{-j\omega}\cdot e^{-j\omega\tau}$
$\to e^{-j\omega - j\omega\tau}$
$\to e^{-j\omega(1+\tau)}$

$d_{B2}(t) \to D_{B2}(\omega)$

$D_{B2}(\omega) \to e^{-j\omega(1+\tau)}$

$D(\omega) = D_A(\omega) + D_{B1}(\omega) + D_{B2}(\omega)$
$= P_\tau(\omega)\cdot e^{-j\omega\tau/2}\cdot\frac{1}{j\omega} + P_\tau(\omega)\cdot e^{-j\omega} + e^{-j\omega(1+\tau)}$
$= P_\tau(\omega)\cdot e^{-j\omega\tau/2}\cdot\frac{1}{j\omega} + P_\tau(\omega)\cdot e^{-j\omega} + e^{-j\omega(1+\tau)}$
$= P_\tau(\omega)\left[e^{-j\omega\tau/2}\cdot\frac{1}{j\omega} + e^{-j\omega}\right] + e^{-j\omega(1+\tau)}$
$= \cancel{\tau}^{1}\operatorname{sinc}\left(\frac{\cancel{\tau}\omega}{2\pi}\right)\left[e^{-j\omega\cancel{\tau}/2}\cdot\frac{1}{j\omega} + e^{-j\omega}\right] + e^{-j\omega(1+\tau)}$, where $\tau = 1$

$D(\omega) = \operatorname{sinc}\left(\frac{\omega}{2\pi}\right)\left[e^{-j\omega/2}\cdot\frac{1}{j\omega} + e^{-j\omega}\right] + e^{-2j\omega}$

BIBO

- $\int_{-\infty}^{\infty} |h(\tau)|\, d\tau < \infty$
- BIBO stability: impulse response is absolutely integrable
- Asymptotic: $t \to \infty$, all modes $e^{\lambda_i t} \to 0$
  - ↳ stable: All real: $\lambda_i < 0$
  - ↳ unstable: at least 1 Real $\lambda_i > 0$ OR repeated roots on imaginary axis
    - ↳ marginally: unrepeated roots on imaginary axis

Identities

- $2\cos\theta = e^{j\theta} + e^{-j\theta}$
- $2j\sin\theta = e^{j\theta} - e^{-j\theta}$
- $j = e^{j\pi/2}$

Discrete Time Systems

- $\mathcal{E} = \sum_{n=-\infty}^{\infty} |x[n]|^2$
- $P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x[n]|^2$
- difference eq. for discrete LTI systems

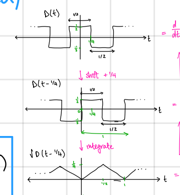
$y[n+N] + a_1 y[n+N-1] + \dots a_{N-1} y[n+1] + a_N y[n] = b_{N-M}\, x[n+M] + b_{N-M+1}\, x[n+M-1] + \dots b_{N-1}\, x[n+1] + b_N\, x[n]$

↳ for causal: $M \leq N$
↳ iterative solution: $y[n] = b_0 x[n] + b_1 x[n-1] + \dots b_N x[n-N] - a_1 y[n-1] - a_2 y[n-2] - \dots a_N y[n-N]$

2024

1a) $h(t) = e^{-t}u(t-1)$
$x(t) = u(t+2)$
$y(t) = h(t) * x(t)$
$= \int h(\tau)\cdot x(t-\tau)$
$= \int_{-\infty}^{\infty} e^{-\tau}u(\tau-1)\cdot u(t-\tau+2)$
$= \int_1^{t+2} e^{-\tau}\cdot 1\cdot 1$
$= \int_1^{t+2} e^{-\tau}$
↓ integrate
$= \int e^{-\tau}\,d\tau$
$= -1\cdot e^{-\tau}$
$= -e^{-\tau}$
↓ w/ bounds
$= -e^{-(t+2)} - -e^{-(1)}$
$= -e^{-(t+2)} + e^{-1}$
$= -e^{-(t+2)} + \frac{1}{e}$
$= \frac{1}{e} - e^{-(t+2)}$

2a) $D(t) \rightarrow D_n = \frac{1}{n\pi}\sin(n\pi/2)$



$= \frac{d}{dt}\mathcal{E}(t+1/4)$
↑ shift −1/4
$= \frac{d}{dt}\mathcal{E}(t)$
↑ differentiate
$= \mathcal{E}(t)$

$D(t) = \frac{d}{dt}\mathcal{E}(t+1/4)$
$\mathcal{E}(t) = \frac{1}{T_0}\int_{T_0}\mathcal{E}(t)\cdot e^{-jn\omega t}\,dt$
$\mathcal{E}(t+1/4) = \frac{1}{T_0}\int_{T_0}\mathcal{E}(t)\cdot e^{-jn\omega(t+1/4)}\,dt$
$= \frac{1}{T_0}\int_{T_0}\mathcal{E}(t)\cdot e^{-jn\omega t}\cdot e^{+jn\omega(1/4)}\,dt$
$\underbrace{\frac{d}{dt}\mathcal{E}(t+1/4)}_{D_n} = \underbrace{\frac{1}{T_0}\int\mathcal{E}(t)\cdot e^{-jn\omega t}}_{\mathcal{E}_n}\cdot\underbrace{e^{+jn\omega(1/4)}\cdot jn\omega}_{\text{to apply to } D_n}$
$D_n = \mathcal{E}_n\cdot e^{jn\omega(1/4)}\cdot jn\omega$
↓
$\mathcal{E}_n = D_n\cdot e^{-jn\omega(1/4)}\cdot\frac{1}{jn\omega}$
$= D_n\cdot e^{-jn2\pi/4}\cdot\frac{1}{jn2\pi}$
where $\omega = 2\pi f = 2\pi(1) = 2\pi$
$= \frac{1}{n\pi}\cdot\sin(n\pi/2)\cdot e^{-jn2\pi/4}\cdot\frac{1}{jn2\pi}$
$= \frac{1}{n\pi\cdot jn2\pi\cdot e^{jn\pi/2}}\cdot\sin(n\pi/2)$
$\mathcal{E}_n = \frac{1}{2jn^2\pi^2}\cdot\sin(n\pi/2)$

$\mathcal{E}_0$ = dc value
= average
$= \frac{1}{T_0}\int_{T_0}\mathcal{E}(t)\,dt$
$= \frac{1}{2}\cdot b\cdot h$
$= \frac{1}{2}\cdot 1\cdot\frac{1}{4}$
$= \frac{1}{8}$

$\therefore\ \mathcal{E}_n = \frac{1}{2jn^2\pi^2}\sin(n\pi/2)$
$\mathcal{E}_0 = 1/8$

5) $H(z) = \frac{1}{z^4(z^2-1/4)}$
$= \frac{1}{z^4(z+1/2)(z-1/2)}$
$= \frac{1}{z^4}\cdot\frac{1}{(z+1/2)(z-1/2)}$

$\frac{1}{(z+1/2)(z-1/2)} = \frac{A}{(z+1/2)} + \frac{B}{(z-1/2)}$
$1 = A(z-1/2) + B(z+1/2)$
$= A(z-1/2) + B(z+1/2)$
$= Az - 1/2A + Bz + 1/2B$
$1 + z0 = \underbrace{z(A+B)}_{=z0} + \underbrace{1/2(-A+B)}_{=1}$
↓
$z0 = z(A+B)$
$0 = A+B$
$A = -B$
$1 = 1/2(-A+B)$
$2 = -A+B$
↓
$2 = -(-B)+B$
$2 = +B+B$
$2 = 2B$
$1 = B$
↓
$A = -1$

$\frac{1}{(z+1/2)(z-1/2)} = \frac{-1}{(z+1/2)} + \frac{1}{(z-1/2)}$
$H(z) = \frac{1}{z^4}\cdot\left[\frac{-1}{z+1/2} + \frac{1}{z-1/2}\right]$
$H(z) = z^{-4}\left[-1\cdot\frac{1}{z-(\frac{-1}{2})} + \frac{1}{z-\frac{1}{2}}\right]$
$h[n] = \delta[n-4]\cdot\left[-1\cdot\left(\frac{-1}{2}\right)^{n-1}u[n-1] + \left(\frac{1}{2}\right)^{n-1}u[n-1]\right]$
$\therefore h[n] = \delta[n-4]\cdot\left[-\left(\frac{-1}{2}\right)^{n-1}u[n-1] + \left(\frac{1}{2}\right)^{n-1}u[n-1]\right]$
?

2023

2a)
$f'(t) = 3\delta(t) - 3\delta(t-2)$
$D_n' = \frac{1}{3}\int_0^3\left[3\delta(t) - 3\delta(t-2)\right]e^{-jn\omega_0 t}\,dt$
$= \frac{1}{3}\int_0^2 3\delta(t)\cdot e^{-jn\omega_0 t} - \frac{1}{3}\int_2^3 3\delta(t-2)\cdot e^{-jn\omega_0 t}\,dt$
$= \cancel{\frac{1}{3}}\cancel{3}\cdot\cancel{e^{-jn\omega 0}} - \cancel{\frac{1}{3}}\cancel{3}\cdot e^{-jn\omega_0(2)}$
$D_n' = 1 - e^{-2jn\omega_0}$
$= 1 - e^{-2jn\frac{2}{3}\pi}$
$= 1 - e^{-4/3\cdot jn\pi}$
$= e^{-4/6jn\pi}\left(e^{4/6jn\pi} - e^{-4/6jn\pi}\right)$
$= e^{-4/6jn\pi}\left(2j\sin(\frac{2}{3}n\pi)\right)$

$\omega_0 = 2\pi f$
$= 2\pi\frac{1}{T_0}$
$= 2\pi\frac{1}{3}$
$= \frac{2}{3}\pi$

$f'(t) = \sum_{n=-\infty}^{\infty} D'(n)\cdot e^{jn\omega_0 t}$
$f(t) = \sum\int e^{-4/6jn\pi}\left(2j\sin(\frac{2}{3}n\pi)\right)\cdot e^{jn\omega_0 t}\,dt$
$= \sum e^{-4/6jn\pi}\left(2j\sin(\frac{2}{3}n\pi)\right)\cdot\int e^{jn\omega_0 t}\,dt$
$= \sum e^{-4/6jn\pi}\left(2j\sin(\frac{2}{3}n\pi)\right)\cdot\frac{1}{jn\omega_0}\,e^{jn\omega_0 t}$
$= \sum e^{-4/6jn\pi}\left(\cancel{2j}\sin(\frac{2}{3}n\pi)\right)\cdot\frac{1}{jn\frac{2}{3}\pi}\,e^{jn\frac{2}{3}\pi t}$
$= \sum e^{jn\frac{2}{3}\pi(t-1)}\cdot\sin(\frac{2}{3}n\pi)\cdot\frac{1}{\pi}$

$f(t) = \frac{1}{\pi}\sum e^{jn\frac{2}{3}\pi(t-1)}\cdot\sin(\frac{2}{3}n\pi)$
$f_0 = \frac{1}{T_0}\cdot A$
$= \frac{1}{3}\cdot 3$
$f_0 = 1$
$\therefore f(t) = \frac{1}{\pi}\sum e^{jn\frac{2}{3}\pi(t-1)}\cdot\sin(\frac{2}{3}n\pi)$
$f_0 = 1$

b)
$\omega = \frac{2}{3}\pi$
$H(\omega)$
$= \frac{10\pi}{j\omega + \frac{2}{\sqrt{3}}\pi}$
$= \frac{10\pi}{j(\frac{2}{3}\pi) + \frac{2}{\sqrt{3}}\pi}$
$= \frac{10}{j\frac{2}{3} + \frac{2}{\sqrt{3}}}$
$= 10\cdot\frac{1}{j(\frac{2}{3}) + \frac{2}{\sqrt{3}}}$
$\angle H(\omega)$
$= \tan^{-1}\left(\frac{-\cancel{2}/3}{\cancel{2}/\sqrt{3}}\right)$
$= \tan^{-1}\left(\frac{-1}{3}\cdot\sqrt{3}\right)$
$= \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)$
$= -30°$

5)
$n \geq 0$: ↳ $u[n]$
$y[n-1] + 2y[n] = x[n]$
$y[-1] = 2$
$x[n] = \left(\frac{1}{4}\right)^n u[n]$

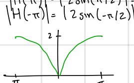
$x[n] \rightarrow X[z]$
$\left(\frac{1}{4}\right)^n u[n] \rightarrow \frac{z}{z-1/4}$
$y[n] \rightarrow Y[z]$
$y[n-1] \rightarrow \frac{1}{z}Y[z] + y[-1]$
$y[n-1] + 2y[n] = x[n]$
$\frac{1}{z}Y[z] + y[-1] + 2Y[z] = \frac{z}{z-1/4}$
$Y[z]\left(\frac{1}{z}+2\right) + y[-1] = \frac{z}{z-1/4}$
$\left(\frac{1}{z}+2\right)Y[z] = \frac{z}{z-1/4} - y[-1]$
$\left(\frac{1}{z}+\frac{2z}{z}\right)Y[z] = \frac{z}{z-1/4} - 2\left(\frac{z-1/4}{z-1/4}\right)$
$\left(\frac{1+2z}{z}\right)Y[z] = \frac{z-2(z-1/4)}{z-1/4}$
$\left(\frac{\cancel{1+2z}}{1}\right)\left(\frac{1}{z}\right)Y[z] = \frac{z-2z+1/2}{z-1/4}\cdot\frac{1}{1+2z}$
$= \frac{-z+1/2}{z-1/4}\cdot\frac{1}{1+2z}$
$= \frac{-z+1/2}{(z-1/4)(2z+1)}$

partial fractions
$\frac{\frac{1}{2}-z}{(z-1/4)(2z+1)} = \frac{A}{z-1/4} + \frac{B}{2z+1}$
$\frac{1}{2} - z = A(2z+1) + B(z-1/4)$
$= A2z + A + Bz - B1/4$
$= \underbrace{z(2A+B)}_{-1z} + \underbrace{A - 1/4B}_{1/2}$
↓
$-1 = 2A+B$
↓
$-1 = 2(\frac{1}{2}+\frac{1}{4}B)+B$
$-1 = 1 + \frac{1}{2}B + B$
$-2 = \frac{3}{2}B$
$-4 = 3B$
$B = \frac{-4}{3}$
$\frac{1}{2} = A - 1/4B$
$A = \frac{1}{2} + 1/4B$
↓
$A = \frac{1}{2} + \frac{1}{4}\cdot\frac{-4}{3}$
$= \frac{1}{2} - \frac{1}{3}$
$= \frac{1}{6}$

$\frac{1}{z}Y[z] = \frac{1/6}{z-1/4} + \frac{-4/3}{2z+1}$
$Y[z] = \frac{1/6\,z}{z-1/4} - \frac{4/3\,z}{2(z+\frac{1}{2})}$
$= \frac{1}{6}\cdot\frac{z}{z-1/4} - \frac{4}{6}\,\frac{z}{z+1/2}$
$y[n] = \left(\frac{1}{6}\right)\left(\frac{1}{4}\right)^n u[n] - \left(\frac{2}{3}\right)\left(\frac{-1}{2}\right)^n u[n]$
$y[n] = \left[\frac{1}{6}\left(\frac{1}{4}\right)^n - \frac{2}{3}\left(\frac{-1}{2}\right)^n\right]u[n]$

4) a)
$y[n] \rightarrow Y(\Omega)$
$x[n] \rightarrow X(\Omega)$
$x[n] + x[n-1] \rightarrow X(\Omega)\cdot e^{-j\Omega\cancel{(1)}}$
$Y(\Omega) = X(\Omega) + X(\Omega)\cdot e^{-j\Omega}$
$= X(\Omega)\left[1 + e^{-j\Omega}\right]$
$H(\Omega) = \frac{Y(\Omega)}{X(\Omega)}$
$= \frac{\cancel{X(\Omega)}\left[1+e^{-j\Omega}\right]}{\cancel{X(\Omega)}}$
$= 1 + e^{-j\Omega}$
$= 1 + e^{-j\Omega}$
$= e^{-j\Omega/2}\left(e^{j\Omega/2} - e^{-j\Omega/2}\right)$
$= e^{-j\Omega/2}\left(2j\cdot\sin(\Omega/2)\right)$
$= 2j\sin(\Omega/2)\cdot e^{-j\Omega/2}$
$= 2\sin(\Omega/2)\cdot e^{-j\Omega/2}\,e^{j\pi/2}$
$= 2\sin(\Omega/2)\cdot e^{j\pi/2 - j\Omega/2}$
$= 2\sin(\Omega/2)\cdot e^{j(\pi-\Omega)/2}$

b) $|H(\Omega)| = |2j\sin(\Omega/2)|$
$= |2\sin(\Omega/2)|$
$|H(0)| = |2\sin(0/2)| = 0$
$|H(\pi)| = |2\sin(\pi/2)| = 2$
$|H(-\pi)| = |2\sin(-\pi/2)| = 2$



c) $\theta(\Omega) = (\pi - \Omega)/2$
$\theta(\Omega) = \begin{cases}\frac{-1}{2}(\pi-\Omega), & 0 \leq \Omega \leq \pi\\ \frac{-1}{2}(\Omega-\pi), & -\pi \leq \Omega \leq 0\end{cases}$

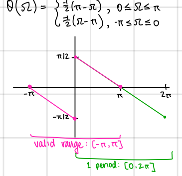


Table of Properties & Pairs

TABLE 5.2 z-Transform Properties

Operation	$x[n]$	$X[z]$
Addition	$x_1[n]+x_2[n]$	$X_1[z]+X_2[z]$
Scalar multiplication	$ax[n]$	$aX[z]$
Right shifting	$x[n-m]u[n-m]$	$\frac{1}{z^m}X[z]$
	$x[n-m]u[n]$	$\frac{1}{z^m}X[z]+\frac{1}{z^m}\sum_{n=1}^{m}x[-n]z^n$
	$x[n-1]u[n]$	$\frac{1}{z}X[z]+x[-1]$
	$x[n-2]u[n]$	$\frac{1}{z^2}X[z]+\frac{1}{z}x[-1]+x[-2]$
	$x[n-3]u[n]$	$\frac{1}{z^3}X[z]+\frac{1}{z^2}x[-1]+\frac{1}{z}x[-2]+x[-3]$
Left shifting	$x[n+m]u[n]$	$z^mX[z]-z^m\sum_{n=0}^{m-1}x[n]z^{-n}$
	$x[n+1]u[n]$	$zX[z]-zx[0]$
	$x[n+2]u[n]$	$z^2X[z]-z^2x[0]-zx[1]$
	$x[n+3]u[n]$	$z^3X[z]-z^3x[0]-z^2x[1]-zx[2]$
Multiplication by γ^n	$\gamma^n x[n]u[n]$	$X\left[\frac{z}{\gamma}\right]$
Multiplication by n	$nx[n]u[n]$	$-z\frac{d}{dz}X[z]$
Time reversal	$x[-n]$	$X[1/z]$
Time convolution	$x_1[n]*x_2[n]$	$X_1[z]X_2[z]$
Initial value	$x[0]$	$\lim_{z\to\infty}X[z]$
Final value	$\lim_{N\to\infty}x[N]$	$\lim_{z\to 1}(z-1)X[z]$ Poles of $(z-1)X[z]$ inside the unit circle

TABLE 6.1 Fourier Series Representation of a Periodic Signal of Period T_0 ($\omega_0 = 2\pi/T_0$)

Series Form	Coefficient Computation	Conversion Formulas
Trigonometric	$a_0 = \frac{1}{T_0}\int_{T_0}f(t)\,dt$	$a_0 = C_0 = D_0$
$f(t) = a_0+\sum_{n=1}^{\infty}a_n\cos n\omega_0t+b_n\sin n\omega_0t$	$a_n = \frac{2}{T_0}\int_{T_0}f(t)\cos n\omega_0t\,dt$	$a_n-jb_n = C_ne^{j\theta_n} = 2D_n$
	$b_n = \frac{2}{T_0}\int_{T_0}f(t)\sin n\omega_0t\,dt$	$a_n+jb_n = C_ne^{-j\theta_n} = 2D_{-n}$
Compact trigonometric	$C_0 = a_0$	$C_0 = D_0$
$f(t) = C_0+\sum_{n=1}^{\infty}C_n\cos(n\omega_0t+\theta_n)$	$\lvert f(t)\rvert = C_n = \sqrt{a_n^2+b_n^2}$	$C_n = 2\lvert D_n\rvert \quad n\geq 1$
	$\theta(f(t)) = \theta_n = \tan^{-1}\left(\frac{-b_n}{a_n}\right)$	$\theta_n = \angle D_n$
Exponential		
$f(t) = \sum_{n=-\infty}^{\infty}D_ne^{jn\omega_0t}$	$D_n = \frac{1}{T_0}\int_{T_0}f(t)e^{-jn\omega_0t}\,dt$	

TABLE 6.2 Selected Fourier Series Properties

Operation	$x(t)$	D_n
Scalar multiplication	$kx(t)$	kD_n
Addition	$x_1(t)+x_2(t)$	$D_{1,n}+D_{2,n}$
	$x_1(t)$, $x_2(t)$ require same ω_0	
Conjugation	$x^*(t)$	D^*_{-n}
Reversal	$x(-t)$	D_{-n}
Time shifting	$x(t-t_0)$	$D_ne^{-jn\omega_0t_0}$
Frequency shifting	$x(t)e^{jn_0\omega_0t}$	D_{n-n_0}
Frequency convolution	$x_1(t)x_2(t)$	$D_{1,n}*D_{2,n}$
	$x_1(t)$, $x_2(t)$ require same ω_0	
Time differentiation	$\frac{d^kx(t)}{dt^k}$	$(jn\omega_0)^kD_n$

TABLE 7.2 Fourier Transform Properties

Operation	$x(t)$	$X(\omega)$
Scalar multiplication	$kx(t)$	$kX(\omega)$
Addition	$x_1(t)+x_2(t)$	$X_1(\omega)+X_2(\omega)$
Conjugation	$x^*(t)$	$X^*(-\omega)$
Duality	$X(t)$	$2\pi x(-\omega)$
Scaling (a real)	$x(at)$	$\frac{1}{\lvert a\rvert}X\left(\frac{\omega}{a}\right)$
Time shifting	$x(t-t_0)$	$X(\omega)e^{-j\omega t_0}$
Frequency shifting (ω_0 real)	$x(t)e^{j\omega_0t}$	$X(\omega-\omega_0)$
Time convolution	$x_1(t)*x_2(t)$	$X_1(\omega)X_2(\omega)$
Frequency convolution	$x_1(t)x_2(t)$	$\frac{1}{2\pi}X_1(\omega)*X_2(\omega)$
Time differentiation	$\frac{d^nx(t)}{dt^n}$	$(j\omega)^nX(\omega)$
Time integration	$\int_{-\infty}^{t}x(u)\,du$	$\frac{X(\omega)}{j\omega}+\pi X(0)\delta(\omega)$

TABLE 5.1 Select (Unilateral) z-Transform Pairs

No.	$x[n]$	$X[z]$
1	$\delta[n-k]$	z^{-k}
2	$u[n]$	$\frac{z}{z-1}$
3	$nu[n]$	$\frac{z}{(z-1)^2}$
4	$n^2u[n]$	$\frac{z(z+1)}{(z-1)^3}$
5	$n^3u[n]$	$\frac{z(z^2+4z+1)}{(z-1)^4}$
6	$\gamma^nu[n]$	$\frac{z}{z-\gamma}$
7	$\gamma^{n-1}u[n-1]$	$\frac{1}{z-\gamma}$
8	$n\gamma^nu[n]$	$\frac{\gamma z}{(z-\gamma)^2}$
9	$n^2\gamma^nu[n]$	$\frac{\gamma z(z+\gamma)}{(z-\gamma)^3}$
10	$\frac{n(n-1)(n-2)\cdots(n-m+1)}{\gamma^mm!}\gamma^nu[n]$	$\frac{z}{(z-\gamma)^{m+1}}$
11a	$\lvert\gamma\rvert^n\cos\beta n\,u[n]$	$\frac{z(z-\lvert\gamma\rvert\cos\beta)}{z^2-(2\lvert\gamma\rvert\cos\beta)z+\lvert\gamma\rvert^2}$
11b	$\lvert\gamma\rvert^n\sin\beta n\,u[n]$	$\frac{z\lvert\gamma\rvert\sin\beta}{z^2-(2\lvert\gamma\rvert\cos\beta)z+\lvert\gamma\rvert^2}$
12a	$r\lvert\gamma\rvert^n\cos(\beta n+\theta)u[n]$	$\frac{rz[z\cos\theta-\lvert\gamma\rvert\cos(\beta-\theta)]}{z^2-(2\lvert\gamma\rvert\cos\beta)z+\lvert\gamma\rvert^2}$
12b	$r\lvert\gamma\rvert^n\cos(\beta n+\theta)u[n]$ $\quad\gamma=\lvert\gamma\rvert e^{j\beta}$	$\frac{(0.5re^{j\theta})z}{z-\gamma}+\frac{(0.5re^{-j\theta})z}{z-\gamma^*}$
12c	$r\lvert\gamma\rvert^n\cos(\beta n+\theta)u[n]$	$\frac{z(Az+B)}{z^2+2az+\lvert\gamma\rvert^2}$

$r = \sqrt{\frac{A^2\lvert\gamma\rvert^2+B^2-2AaB}{\lvert\gamma\rvert^2-a^2}}$

$\beta = \cos^{-1}\frac{-a}{\lvert\gamma\rvert}$

$\theta = \tan^{-1}\frac{Aa-B}{A\sqrt{\lvert\gamma\rvert^2-a^2}}$

TABLE 7.1 Select Fourier Transform Pairs

No.	$x(t)$	$X(\omega)$	
1	$e^{-at}u(t)$	$\frac{1}{a+j\omega}$	$a>0$
2	$e^{at}u(-t)$	$\frac{1}{a-j\omega}$	$a>0$
3	$e^{-a\lvert t\rvert}$	$\frac{2a}{a^2+\omega^2}$	$a>0$
4	$te^{-at}u(t)$	$\frac{1}{(a+j\omega)^2}$	$a>0$
5	$t^ne^{-at}u(t)$	$\frac{n!}{(a+j\omega)^{n+1}}$	$a>0$
6	$\delta(t)$	1	
7	1	$2\pi\delta(\omega)$	
8	$e^{j\omega_0t}$	$2\pi\delta(\omega-\omega_0)$	
9	$\cos\omega_0t$	$\pi[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$	
10	$\sin\omega_0t$	$j\pi[\delta(\omega+\omega_0)-\delta(\omega-\omega_0)]$	
11	$u(t)$	$\pi\delta(\omega)+\frac{1}{j\omega}$	
12	$\operatorname{sgn}t$	$\frac{2}{j\omega}$	
13	$\cos\omega_0t\,u(t)$	$\frac{\pi}{2}[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]+\frac{j\omega}{\omega_0^2-\omega^2}$	
14	$\sin\omega_0t\,u(t)$	$\frac{\pi}{2j}[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]+\frac{\omega_0}{\omega_0^2-\omega^2}$	
15	$e^{-at}\sin\omega_0t\,u(t)$	$\frac{\omega_0}{(a+j\omega)^2+\omega_0^2}$	$a>0$
16	$e^{-at}\cos\omega_0t\,u(t)$	$\frac{a+j\omega}{(a+j\omega)^2+\omega_0^2}$	$a>0$
17	$\operatorname{rect}\left(\frac{t}{\tau}\right)$	$\tau\operatorname{sinc}\left(\frac{\omega\tau}{2}\right)$	
18	$\frac{W}{\pi}\operatorname{sinc}(Wt)$	$\operatorname{rect}\left(\frac{\omega}{2W}\right)$	
19	$\Delta\left(\frac{t}{\tau}\right)$	$\frac{\tau}{2}\operatorname{sinc}^2\left(\frac{\omega\tau}{4}\right)$	
20	$\frac{W}{2\pi}\operatorname{sinc}^2\left(\frac{Wt}{2}\right)$	$\Delta\left(\frac{\omega}{2W}\right)$	
21	$\sum_{n=-\infty}^{\infty}\delta(t-nT)$	$\omega_0\sum_{n=-\infty}^{\infty}\delta(\omega-n\omega_0)$	$\omega_0=\frac{2\pi}{T}$
22	$e^{-t^2/2\sigma^2}$	$\sigma\sqrt{2\pi}e^{-\sigma^2\omega^2/2}$	

TABLE 9.1 Select Discrete-Time Fourier Transform Pairs

No.	$x[n]$	$X(\Omega)$	
1	$\delta[n-k]$	$e^{-jk\Omega}$	Integer k
2	$\gamma^nu[n]$	$\frac{e^{j\Omega}}{e^{j\Omega}-\gamma}$	$\lvert\gamma\rvert<1$
3	$-\gamma^nu[-(n+1)]$	$\frac{e^{j\Omega}}{e^{j\Omega}-\gamma}$	$\lvert\gamma\rvert>1$
4	$\gamma^{\lvert n\rvert}$	$\frac{1-\gamma^2}{1-2\gamma\cos\Omega+\gamma^2}$	$\lvert\gamma\rvert<1$
5	$n\gamma^nu[n]$	$\frac{\gamma e^{j\Omega}}{(e^{j\Omega}-\gamma)^2}$	$\lvert\gamma\rvert<1$
6	$\gamma^n\cos(\Omega_0n+\theta)u[n]$	$\frac{e^{j\Omega}[e^{j\Omega}\cos\theta-\gamma\cos(\Omega_0-\theta)]}{e^{j2\Omega}-(2\gamma\cos\Omega_0)e^{j\Omega}+\gamma^2}$	$\lvert\gamma\rvert<1$
7	$u[n]-u[n-M]$	$\frac{\sin(M\Omega/2)}{\sin(\Omega/2)}e^{-j\Omega(M-1)/2}$	
8	$\frac{\Omega_c}{\pi}\operatorname{sinc}(\Omega_cn)$	$\sum_{k=-\infty}^{\infty}\operatorname{rect}\left(\frac{\Omega-2\pi k}{2\Omega_c}\right)$	$\Omega_c\leq\pi$
9	$\frac{\Omega_c}{2\pi}\operatorname{sinc}^2\left(\frac{\Omega_cn}{2}\right)$	$\sum_{k=-\infty}^{\infty}\Delta\left(\frac{\Omega-2\pi k}{2\Omega_c}\right)$	$\Omega_c\leq\pi$
10	$u[n]$	$\frac{e^{j\Omega}}{e^{j\Omega}-1}+\pi\sum_{k=-\infty}^{\infty}\delta(\Omega-2\pi k)$	
11	1 for all n	$2\pi\sum_{k=-\infty}^{\infty}\delta(\Omega-2\pi k)$	
12	$e^{j\Omega_0n}$	$2\pi\sum_{k=-\infty}^{\infty}\delta(\Omega-\Omega_0-2\pi k)$	
13	$\cos\Omega_0n$	$\pi\sum_{k=-\infty}^{\infty}\delta(\Omega-\Omega_0-2\pi k)+\delta(\Omega+\Omega_0-2\pi k)$	
14	$\sin\Omega_0n$	$j\pi\sum_{k=-\infty}^{\infty}\delta(\Omega+\Omega_0-2\pi k)-\delta(\Omega-\Omega_0-2\pi k)$	
15	$(\cos\Omega_0n)u[n]$	$\frac{e^{j2\Omega}-e^{j\Omega}\cos\Omega_0}{e^{j2\Omega}-2e^{j\Omega}\cos\Omega_0+1}+\frac{\pi}{2}\sum_{k=-\infty}^{\infty}\delta(\Omega-2\pi k-\Omega_0)+\delta(\Omega-2\pi k+\Omega_0)$	
16	$(\sin\Omega_0n)u[n]$	$\frac{e^{j\Omega}\sin\Omega_0}{e^{j2\Omega}-2e^{j\Omega}\cos\Omega_0+1}+\frac{\pi}{2j}\sum_{k=-\infty}^{\infty}\delta(\Omega-2\pi k-\Omega_0)-\delta(\Omega-2\pi k+\Omega_0)$	